

Semester Overview

Full credit: Mr. Antu Mitra(Real Topper)

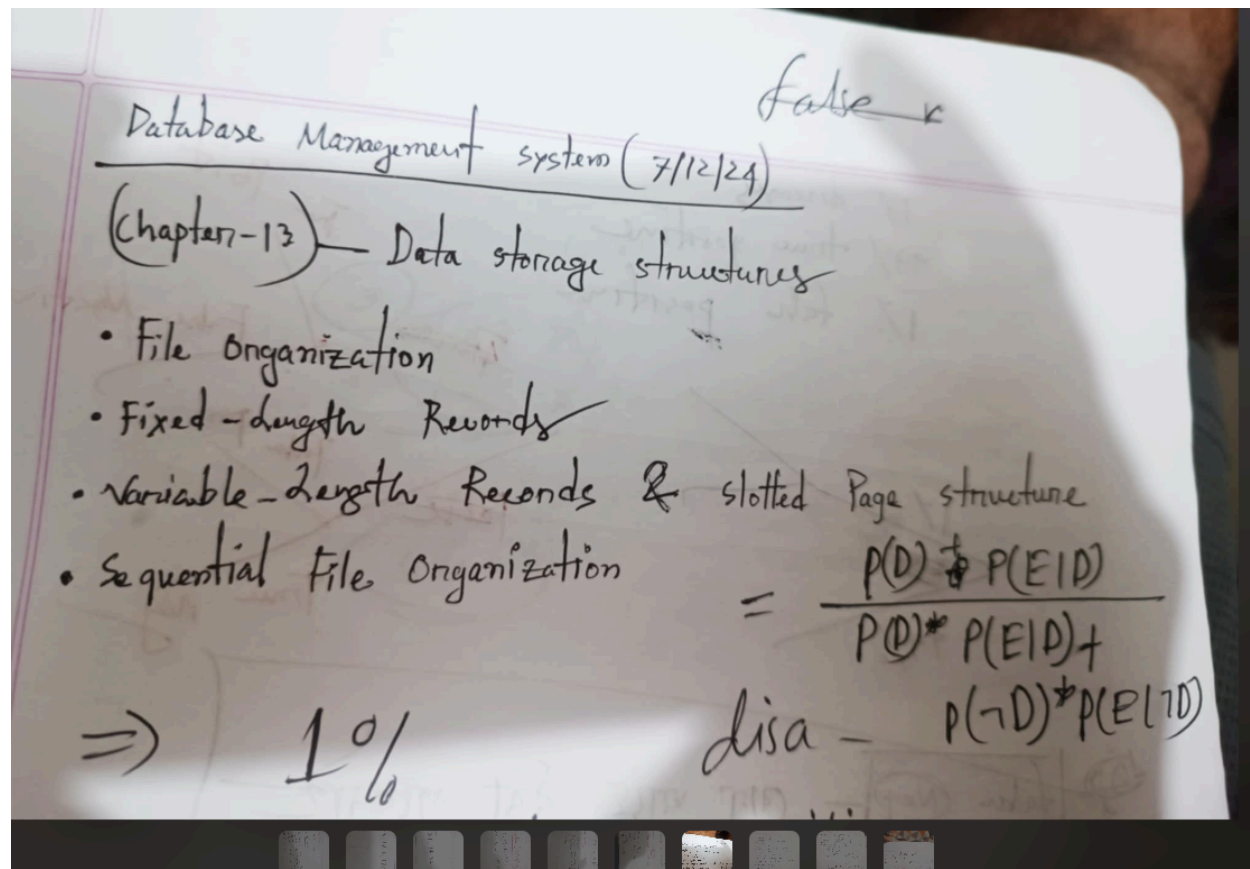
DataBase

Database Management System

Ch-6: Database design using the E-R model

- Design phase
- Avoid two major pitfalls (Design alternatives)
 - Redundancy
 - Incompleteness
- Design Approaches
- Entity set
- Representing Entity sets in Diagram
- Relationship sets
- Representing Relationship sets via ER Diagram
- Relationship sets with attributes
- Degree of a Relationship set
- Ternary Relationship

chap-6



chap-13

9.4 - Design of Arithmetic circuit

Database Management system (10/12/24)

(Chapter-10) Additional — B⁺ Tree

- B⁺ Tree structure of B⁺ Tree
- Searching a record in B⁺ Tree
- B⁺ Tree insertion
- B⁺ Tree Deletion
- Advantages of B⁺ Trees

Quiz — Best

Database Management system (11/01/25)

Topic — 7.25, 7.26, 7.27, 7.28, 7.29
7.30, 7.31, 7.32, 7.39, 7.40
7.41, 7.42, 7.46, 7.52, 7.53
7.72, Chapter-7

Chapter-7 → 1 theke 25 page o portae hbe

Database Management system (13/01/24)

Topics: 14.3, 14.4, 14.5, 14.6, 14.7, 14.8, 14.9, 14.10, *
14.18, 14.19, 14.20, 14.21, 14.22, 14.23, 14.25,

Dates: (14/01/29)
Topic: 14.29, 14.30, 14.31, 14.32, 14.33, 14.34, 14.35,

Chapter-14

Topics: 17.3, 17.4, 17.5, 17.6, 17.7, 17.8, 17.9

Chapter 17

Databox (18/01/25)
Topic: 17.16, 17.11, 17.12
Databox (29/01/25)

Chap-17

Database (19.11.20)

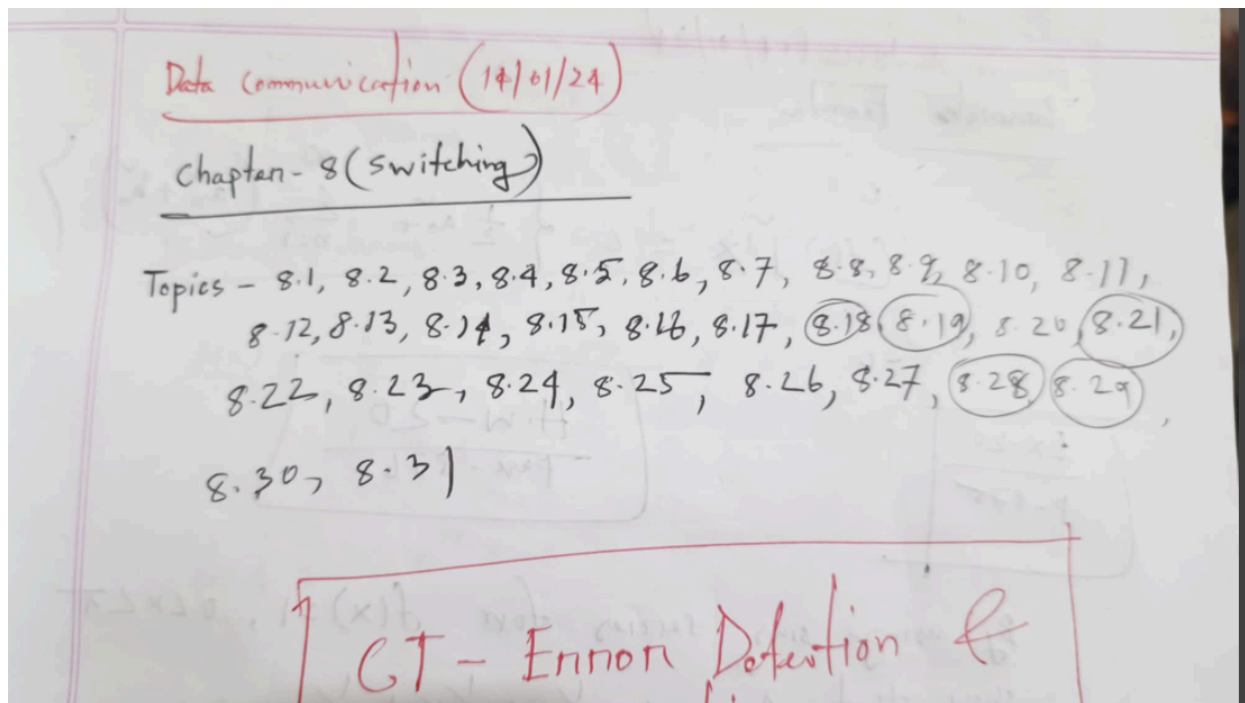
Topics: 19.1, 19.2, 19.3, 19.4, 19.5, 19.6, Chapter-19
19.7, 19.8, 19.9, 19.10, 19.11, 19.12,
19.13, 19.14, 19.15, 19.17, 19.18,
19.19, 19.20, 19.21, 19.22, 19.23,
(19.24 to 19.29) - (GATE)

Chap-19

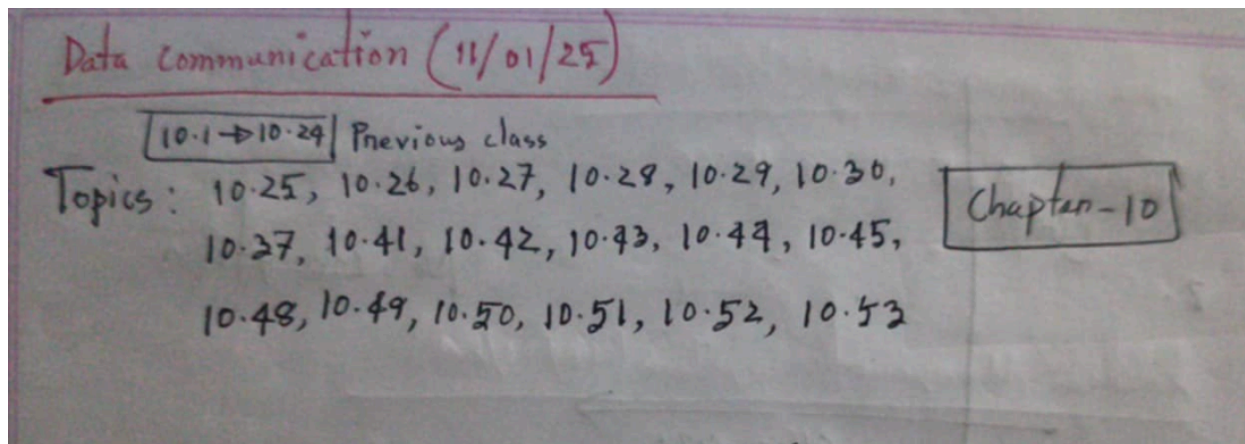
Chap-5 → not clear

Data Communication

Chapter 1 to 7 full

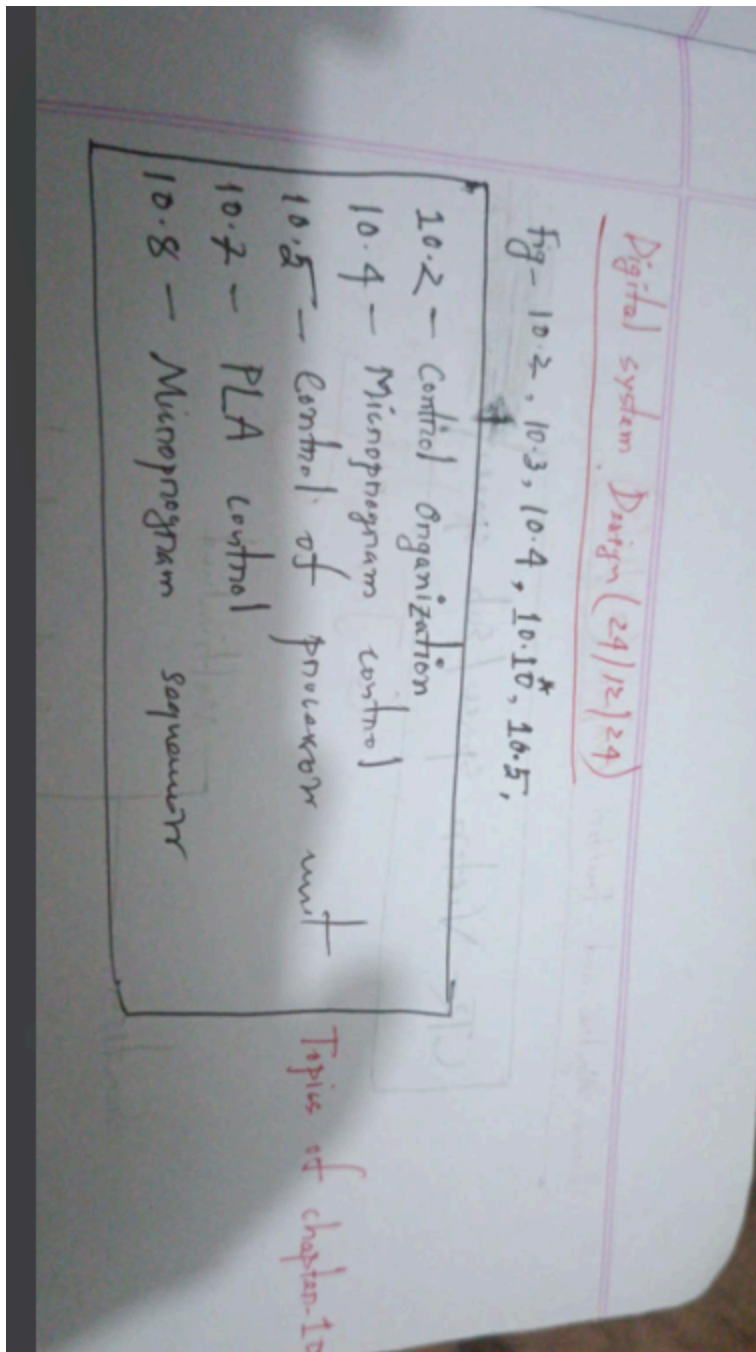


Chapter-10→



DSD

8,9-Full



11-full

12-not clear

Probability

Yesterday				
	Probability_and_Statistics_for_Engineers_...	1/29/2025 10:47 PM	Microsoft Edge PD...	6,165 KB
Last month				
	Discrete Distributions.pptx	12/23/2024 10:41 AM	Microsoft PowerPo...	794 KB
	Event Relations and Probability Rules.pptx	12/23/2024 10:41 AM	Microsoft PowerPo...	633 KB
	Probability.pptx	12/23/2024 10:41 AM	Microsoft PowerPo...	333 KB
	Random_Variables,_Mathematical_Expect...	12/23/2024 10:41 AM	Microsoft PowerPo...	553 KB
	SleepData.xlsx	12/23/2024 10:41 AM	Microsoft Excel W...	7 KB
	Stat.pptx	12/23/2024 10:41 AM	Microsoft PowerPo...	644 KB

All slides on google classroom

In book→

Preface	xv
✓ 1 Introduction to Statistics and Data Analysis	1
1.1 Overview: Statistical Inference, Samples, Populations, and the Role of Probability	1
1.2 Sampling Procedures; Collection of Data	7
1.3 Measures of Location: The Sample Mean and Median	11
Exercises	13
1.4 Measures of Variability	14
Exercises	17
1.5 Discrete and Continuous Data	17
1.6 Statistical Modeling, Scientific Inspection, and Graphical Diagnostics	18
1.7 General Types of Statistical Studies: Designed Experiment, Observational Study, and Retrospective Study	27
Exercises	30
✓ 2 Probability	35
2.1 Sample Space	35
2.2 Events	38
Exercises	42
2.3 Counting Sample Points	44
Exercises	51
2.4 Probability of an Event	52
2.5 Additive Rules	56
Exercises	59
2.6 Conditional Probability, Independence, and the Product Rule ...	62
Exercises	69
2.7 Bayes' Rule	72
Exercises	76
Review Exercises	77

	in Other Chapters.....	79
	3 Random Variables and Probability Distributions	81
	3.1 Concept of a Random Variable	81
	3.2 Discrete Probability Distributions	84
	3.3 Continuous Probability Distributions	87
	Exercises	91
	3.4 Joint Probability Distributions	94
	Exercises	104
	Review Exercises.....	107
	3.5 Potential Misconceptions and Hazards; Relationship to Material in Other Chapters.....	109
	4 Mathematical Expectation	111
	4.1 Mean of a Random Variable.....	111
	Exercises	117
	4.2 Variance and Covariance of Random Variables.....	119
	Exercises	127
	4.3 Means and Variances of Linear Combinations of Random Variables	128
	4.4 Chebyshev's Theorem	135
	Exercises	137
	Review Exercises.....	139
	4.5 Potential Misconceptions and Hazards; Relationship to Material in Other Chapters.....	142
	5 Some Discrete Probability Distributions	143

✓ 11	Simple Linear Regression and Correlation	389
✓ 11.1	Introduction to Linear Regression	389
✓ 11.2	The Simple Linear Regression Model	390
✓ 11.3	Least Squares and the Fitted Model	394
	Exercises	398
11.4	Properties of the Least Squares Estimators	400
11.5	Inferences Concerning the Regression Coefficients	403
11.6	Prediction	408
	Exercises	411
11.7	Choice of a Regression Model	414
11.8	Analysis-of-Variance Approach	414
11.9	Test for Linearity of Regression: Data with Repeated Observations	416
	Exercises	421
11.10	Data Plots and Transformations	424
11.11	Simple Linear Regression Case Study	428
11.12	Correlation	430
	Exercises	435
	Review Exercises	436
11.13	Potential Misconceptions and Hazards; Relationship to Material	

Last class e permutation and combination basic poraisae aro poracche....

Linear algebra and Fourier Analysis

Matrix:

Types of Matrix

1. Identity Matrix — I (कई बराबर)
2. Scalar Matrix — kI (कई बराबर k गुणित)
3. Commute Matrix — square (A & B) एवं $AB = BA$
4. Anticommutate Matrix — square (A & B) एवं $AB = -BA$
5. Periodic — $A^{k+1} = A$ where $k = \text{positive integer}$
6. Idempotent — $A^2 = A$
7. Nilpotent — $A^p = 0$ where $p = \text{positive integer}$
8. Inverse Matrix — $AB = I$
9. Transpose Matrix — A^T
10. Symmetric Matrix — $A^T = A$
11. Skewsymmetric Matrix — $A^T = -A$
12. Orthogonal Matrix — $AA^T = A^T A = I$
13. Involutary Matrix — $A^2 = I$
14. Singular Matrix — $|A| \neq 0$
15. Conjugate Matrix —

$$A = \begin{bmatrix} 1+2i & i \\ 3 & 2-3i \end{bmatrix} \quad \bar{A} = \begin{bmatrix} 1-2i & -i \\ 3 & 2+3i \end{bmatrix}$$

16. Hermitian Matrix — $A^T = \bar{A}$ or $\bar{A}^T = A$

17. Skew Hermitian Matrix — $A^T = -\bar{A}$ or $\bar{A}^T = -A$

18. Normal Matrix — $A^* A \neq I \neq A A^*$

19. Unitary Matrix — $A^* A = I = A A^*$

System of a linear equation of a matrix

$$2x - y = 4$$

$$3x + 2y = 13$$

$$\begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 13 \end{bmatrix}$$

↓
A

↓
X

↓
B

$$AX = B$$

$$X = A^{-1}B$$

Row Echelon form

1. અમુક યાદિવત્ ગુણ યાદિત કરાવે શકાય.
2. અમુક ઉપાધતુઓ leading એપાધતવ હોતે શકે શકાય.
3. ક્રમાંક Leading એપાધતવ નીચે ઉપાધતુઓ અમુક રહે.

Rank of a matrix

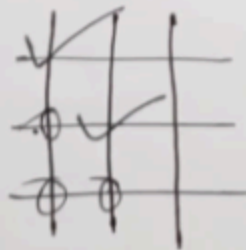
Number of independent row = Rank

Way-1 (determinant)

$$A = 3 \times 3$$

$|A| = 0$ then if one minor is non-zero
then the Rank = 2 but if all the minors
are zero then Rank = 1

Way-2 (Row Echelon)



Inverse of a matrix using row canonical form

- All the work need to be done using the row only.

$$\left(\begin{array}{ccc|ccc} 3 & 4 & -1 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 2 & 5 & -4 & 0 & 0 & 1 \end{array} \right)$$

Inverse of a matrix using column canonical form

- All the work need to be done using the column only.

Transpose and Normal of a matrix

Answer's file

$$\begin{aligned} 2x + y - z &= 1 \\ 3x + 2y + 2z &= 13 \\ 4x - 2y + 3z &= 9 \end{aligned}$$

$$\begin{aligned} a_1x + b_1y + c_1z &= d_1 \\ a_2x + b_2y + c_2z &= d_2 \\ a_3x + b_3y + c_3z &= d_3 \end{aligned}$$

$$X = \frac{D_x}{D}$$

$$Y = \frac{D_y}{D}$$

$$Z = \frac{D_z}{D}$$

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 33$$

$$D_x = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix} = 33 \quad \begin{matrix} x=1 \\ y=2 \\ z=2 \end{matrix}$$

$$D_y = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix} = 66$$

$$D_z = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix} = 99$$

Co-efficient

$$2x + 4y - 6z = 8$$

$$-x - y + 2z = -3$$

$$6x + 2y - z = 0$$

Co-efficient matrix $X =$

$$\begin{bmatrix} 2 & 4 & -6 \\ -1 & -1 & 2 \\ 6 & 2 & -1 \end{bmatrix}$$

Augmentation Matrix

The augmented matrix is a rectangular matrix. The number of columns is equal to the variables in the linear equations and the constant term. The number of rows is equal to the number of linear equations.

$$A = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad B = \begin{bmatrix} 1 & -2 \\ 0 & 4 \end{bmatrix}$$

$$A|B = \begin{bmatrix} 2 & 1 & -2 \\ 3 & 0 & 4 \end{bmatrix}$$

$$3|A = \begin{bmatrix} 1 & -2 & 2 \\ 0 & 4 & 3 \end{bmatrix}$$

Homogeneous / Non-homogeneous

If the value of the variable is zero then it is homogeneous otherwise it is nonhomogeneous.

**SCHAUM'S
OUTLINE
SERIES**

THEORY and PROBLEMS

of
MATRICES

by **FRANK AYRES, JR.**

including
**340
solved
problems**

Ei Topics gula ei boi theke portae paren.

COLLEGE
LINEAR ALGEBRA

THEORY OF MATRICES

PROFESSOR MD. ABDUR RAHAMAN

$$\begin{bmatrix} a_{11} & a_{12} & \dots & \dots & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & \dots & \dots & a_{2n} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & \dots & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

$$\mathbb{R}^3 = \{(a, b, c) \mid a, b, c \in \mathbb{R}\}$$

$$T: U \rightarrow V$$

$$T(u_1 + u_2) = T(u_1) + T(u_2)$$

$$T(\alpha u) = \alpha T(u)$$

$$\alpha \in \mathbb{R} \text{ \& } u_1, u_2, u \in U$$

$$(AB)^+ = B^*(BB^*)^{-1} (A^*A)^{-1} A^*$$

if $(BB^*)^{-1}$ \& $(A^*A)^{-1}$ exist

NAHAR BOOK DEPOT & PUBLICATIONS
BANGLABAZAR, DHAKA

Ei boi er sodho

and hence the system has no solution.

Example 11. For what values of λ and μ the following system of linear equations has (i) no solution

(ii) more than one solution (iii) a unique solution :

$$\begin{cases} x+y+z = 6 \\ x+2y+3z = 10 \\ x+2y+\lambda z = \mu \end{cases} \quad [\text{D. U. H 1980; D. U. H (Stat) 1982}]$$

Solution: Reduce the given system to echelon form by the elementary operations. We subtract 1st equation from 2nd and 3rd equations. Then we have the equivalent system.

$$\begin{cases} x+y+z = 6 \\ y+2z = 4 \\ y+(\lambda-1)z = \mu-6 \end{cases}$$

We subtract 2nd equation from 3rd equation. Then we have the equivalent system

$$\begin{cases} x+y+z = 6 \\ y+2z = 4 \\ (\lambda-3)z = \mu-10 \end{cases} \quad (*)$$

Now from the last system (*) we have the following three cases:

(i) If $\lambda = 3$ and $\mu \neq 10$ then the 3rd equation of (*) is of the form $0 = a$ where $a = \mu - 10 \neq 0$ which implies that 0 is equal to a non-zero real number which is not true. Thus we conclude that for $\lambda = 3$ and $\mu \neq 10$, the given system has no solution.

three unknowns, so it has $3 - 2 = 1$ free variable which is z and hence for $\lambda = 3$ and $\mu = 10$, the given system will have more than one solution.

(iii) For a unique solution the coefficient of z in the 3rd equation must be non-zero i.e. $\lambda \neq 3$ and μ may have any value. Thus for $\lambda \neq 3$ and μ arbitrary the given system have a unique solution.

Example 12. For what values of λ , the following linear equations have a solution and solve them completely in each case :

$$\begin{cases} x+y+z = 1 \\ x+2y+4z = \lambda \\ x+4y+10z = \lambda^2 \end{cases}$$

Solution: The given system of linear equations is

$$\begin{cases} x+y+z = 1 \\ x+2y+4z = \lambda \\ x+4y+10z = \lambda^2 \end{cases}$$

Reduce the given system to echelon form by the elementary operations. We subtract 1st equation from 2nd & 3rd equations respectively. Then we have the equivalent system.

$$\begin{cases} x+y+z = 1 \\ y+3z = \lambda-1 \\ 3y+9z = \lambda^2-1 \end{cases}$$

We multiply 2nd equation by 3 and then subtract from the 3rd equation. Then we have the equivalent system

$$\begin{cases} x+y+z = 1 \\ y+3z = \lambda-1 \\ 0+0 = \lambda^2-3\lambda+2 \end{cases} \Rightarrow \begin{cases} x+y+z = 1 \\ y+3z = \lambda-1 \\ 0 = \lambda^2-3\lambda+2 \end{cases} \quad (*)$$

Page 26 er Example: 11,12 important(12 ct tae disilo)
+ Class er korano math

Vector:



Linear Algebra

Fourth Edition

612 fully solved problems

- Concise explanations of all course concepts
- Information on algebraic systems, polynomials, and matrix applications

Ei boi er chapter 4

4.1-4.5 er sokol example + oi related math

Class e korano math

Let V be a vector space over a field K and
let $v_1, v_2, \dots, v_n \in V$. Any vector in of the form
 $a_1 v_1 + a_2 v_2 + \dots + a_n v_n$.

where $a_i \in K$, is called a linear combination of
 $v_1, v_2, \dots, v_n$.

N.B

The set of all linear combination of vector
set S , $L(S)$, is a subspace of V containing
 S .

Ex
Determine whether or not the vector,

$v = (3, 9, -4, -2)$ is a linear combination
of the vectors $u_1 = (1, -2, 0, 3)$ $u_2 = (2, 3, 0, -1)$
 $u_3 = (2, -1, 2, 1)$

$$v = xu_1 + yu_2 + zu_3$$

② Define what is linear combination? Check if this
consistent or not?

$$(3, 9, -4, -2) = x(1, -2, 0, 3) + y(2, 3, 0, -1) + z(2, -1, 2, 1)$$

$$x + 2y + 2z = 3$$

$$-2x + 3y - z = 9$$

$$2z = -4$$

$$3x - y + z = -2$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 2 & 3 \\ -2 & 3 & -1 & 9 \\ 0 & 0 & -4 & -4 \\ 3 & -1 & 1 & -2 \end{array} \right)$$

H.W

$$v = 2t^2 + 4t - 2$$

$$l_1 = t^2 - 2t + 5$$

$$l_2 = 2t^2 - 3t$$

$$l_3 = t + 2$$

$$E = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$$

$$B = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$$

$$C = \begin{pmatrix} 0 & 2 \\ 0 & -1 \end{pmatrix}$$

Tutorialpoint — python website

linux ubuntu — Job

Greeky shows — python

Linear Algebra (17/12/24)

Basis and ~~Dimension~~ Dimension

A vector space V is said to be of finite dimensional if there exists linearly independent vectors $e_1, e_2, \dots, e_n$ which span V . The sequence $\{e_1, e_2, \dots, e_n\}$ is then called a basis of V .

Coordinates

Let $\{e_1, e_2, \dots, e_n\}$ be a basis of an n -dimensional space V over K and let v be any vector in V . Since $\{e_i\}$ generates V , v is a linear combination of the e_i ,

$$v = a_1 e_1 + a_2 e_2 + \dots + a_n e_n$$

$$[v]_e = (a_1, a_2, \dots, a_n)$$

Basis condition

1. Linearly independent
2. Span

Question 1

$(1, 1, 2), (1, 2, 5), (5, 3, 4)$ three vectors can build a $\mathbb{R}^3$ Basis.

=====

Question 2

$(1, 1, 1), (1, 2, 3)$ and $(2, -1, 1)$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & -3 & -1 \end{bmatrix}$$

Rank 3

Question-3

Let W be the subspace of $\mathbb{R}^4$ generated by the vectors $(1, -2, 5, -3)$, $(2, 3, 1, -4)$ and $(3, 8, -3, -5)$.

- (I) Find a basis and dimension of W .
- (II) Extend the basis of W to a basis of the whole space $\mathbb{R}^4$.

Answer

$$I. \begin{bmatrix} 1 & -2 & 5 & -3 \\ 2 & 3 & 1 & -4 \\ 3 & 8 & -3 & -5 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -2 & 5 & -3 \\ 0 & 7 & -9 & 2 \\ 0 & 14 & -18 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -2 & 5 & -3 \\ 0 & 7 & -9 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{Rank} = 2$$

so the basis of W

$$\left\{ (1, -2, 5, -3), (0, 7, -9, 2) \right\}$$

Rank = Dimension

Dimension of $W = 2$

II.

Question 4
Find the dimension and a basis of the solution space W of the system

$$\begin{aligned}x + 2y + 2z - s + 3t &= 0 \\x + 2y + 3z + s + t &= 0 \\3x + 6y + 8z + s + 5t &= 0\end{aligned}$$

$$\left[\begin{array}{ccccc|ccc} 1 & 2 & 2 & -1 & 3 & 1 & 0 & 0 \\ 1 & 2 & 3 & 1 & 1 & 1 & 0 & 0 \\ 3 & 6 & 8 & 1 & 5 & 1 & 0 & 0 \end{array} \right]$$

Add an augmentation

$$\begin{aligned}x + 2y + 2z - s + 3t &= 0 \\z + 2s - 2t &= 6\end{aligned}$$

Dimension = Free Variable

Set 1:

$$y = 1, s = 1, t = 1$$

Set 2:

Set 3:

$$\begin{aligned}z &= 0 \\x &= -4\end{aligned}$$

$$(-4, 1, 0, 1, 1)$$

Question-5

Let V and W be the subspace of $\mathbb{R}^3$.

$$V = \{(a, b, c) : b+c=0\}$$

$$W = \{(a, b, c) : a+b=0, c=2d\}$$

Find the basis and dimension of

(i) V

(ii) W

(iii) $V \cap W$

Ans

i. V is Dimension = 2

ii. W is Dimension = 2

iii. $V \cap W$ is Dimension = 1

Basis Using Question is given

$$\dim(U+W) = \dim(U) + \dim(W) - \dim(U \cap W)$$

$$4 = 2 + 2 - 1$$

Question-6

$$V = \{ at^u + bt + c; a, b, c \in \mathbb{R} \}$$

$$V = 2t^u - 5t + 6$$

$$e_1 = 1, e_2 = t-1, e_3 = (t-1)^u$$

$$V = xe_1 + ye_2 + ze_3$$

$$[V]_e = (x, y, z)$$

$$= (3, -1, 2)$$

Fouriar Analysis:

**ADVANCED
ENGINEERING MATHEMATICS**

Ei boi er:

12. Fourier Series

850–884

12.1 Periodic Functions (850); 12.2 Fourier series (850); 12.3 Dirichlet's Conditions (851); 12.4 Advantages of Fourier Series (851); 12.5 Useful Integrals (851); 12.6 Determination of Fourier constants (Euler's Formulae) (851); 12.7 Functions defined in two or more sub spaces (855); 12.8 Even Functions (861); 12.9 Half Range's series (864); 12.10 Change of Interval (866); 12.11 Parseval's Formula (874); 12.12 Fourier series in Complex Form (879); 12.13 Practical Harmonic Analysis (880).

14. Integral Transforms

933–981

14.1 Introduction (933); 14.2 Integral Transforms (933); 14.3 Fourier Integral Theorem (934); 14.4 Fourier sine and cosine Integrals (935); 14.5 Fourier's Complex Integral (936); 14.6 Fourier Transforms (938); 14.7 Fourier sine and cosine Transforms (939); 14.8 Properties of Fourier Transform (947); 14.9 Convolution (951); 14.10 Parseval's Identity for Fourier Transform (951); 14.11 Parseval's identity cosine Transform (952); 14.12 Parseval's identity for sine Transform (952); 14.13 Fourier Transforms of Derivative of a function (958); 14.14 Relationship Between Fourier and Laplace Transforms (959); 14.15 Solution of Boundary value problems by using integral transform (959); 14.16 Fourier Transforms of Partial Derivative of a Function (965); 14.17 Finite Fourier Transforms (969); 14.18 Finite Fourier sine and Cosine transforms of Derivatives (976).

Linear Algebra and Fourier Analysis (29/12/21)

CTR- Vector space / sub space

6 January

function ———— $\left\{ \begin{array}{l} \text{Multivalued} \\ \text{single valued} \end{array} \right.$

odd function ~~is a~~

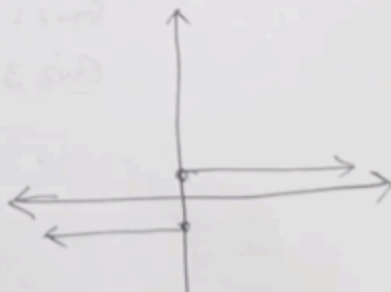
even function

Periodic function

$$y = a \sin(bx + c)$$

continuity

$$y = \frac{|x|}{x}$$



Chapter - 12

$$1. \int_{-a}^a f(x) dx = 0 \quad \text{if } f(x) \text{ is an odd function}$$

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx \quad \text{if } f(x) \text{ is an odd function}$$

$$\begin{aligned} 2. \int_{-\pi}^{\pi} \cos nx \, dx &= \left[\frac{\sin nx}{n} \right]_{-\pi}^{\pi} \\ &= \frac{1}{n} (\sin n\pi + \sin n\pi) \\ &= \frac{2 \sin n\pi}{n} \\ &\approx 0 \end{aligned}$$

$$3. \int_{-\pi}^{\pi} \cos nx \, dx$$

④ A series of $\sin$ and $\cos$ of an angle and its multiples of the form

$$\frac{a_0}{2} + a_1 \cos x + a_2 \cos 2x + \dots + b_1 \sin x + b_2 \sin 2x + \dots$$

$$= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

Dirichlet's condition for a Fourier series
 If the function $f(x)$ for the interval $(-\pi, \pi)$

- (1) single-valued
- (2) bounded
- (3) has at most a finite number of maxima and minima
- (4) has only a finite number of discontinuities
- (5) if $f(x+2\pi) = f(x)$ for values of x outside $[-\pi, \pi]$ then,

$$S_p(x) = \frac{a_0}{2} + \sum_{n=1}^p a_n \cos nx + \sum_{n=1}^p b_n \sin nx$$

converges to $f(x)$ as $p \rightarrow \infty$ at values of x for which $f(x)$ is continuous and to $\frac{1}{2}[f(x+0) + f(x-0)]$ at points of discontinuity.

Fourier Analysis (07/01/25)

Find the Fourier series representing

$$f(x) = x, \quad 0 < x < 2\pi$$

and sketch its graph from $x = -4\pi$ to 4π

$$y = x$$

soln: Let $f(x) = \frac{a_0}{2} + a_1 \cos x + a_2 \cos 2x + \dots + b_1 \sin x + b_2 \sin 2x + \dots$ — (1)

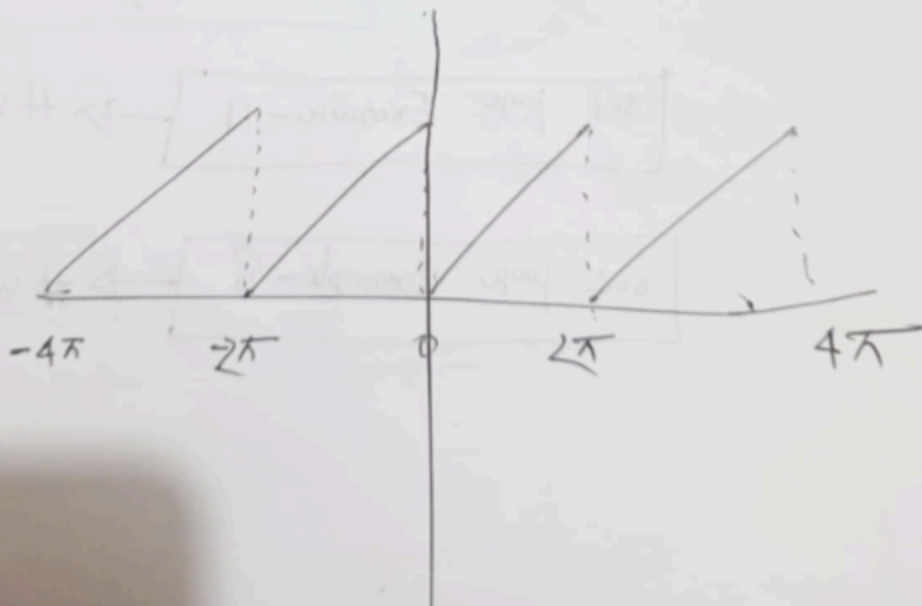
$$a_0 = \frac{1}{\pi} \int_0^{2\pi} x dx = 2\pi$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_0^{\pi} x \cos nx dx = 0$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx = \frac{1}{\pi} \int_0^{\pi} x \sin nx dx = -\frac{2}{n}$$

Now from (1)

$$x = \pi - 2 \left[\sin x + \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x + \dots \right]$$



859 page Exercise-2 $\rightarrow$ H.W

Find the Fourier series representing

$$f(x) = \begin{cases} x^2 & -\pi < x < 0 \\ 0 & 0 < x < \pi \end{cases}$$

$$f(x) = \frac{a_0}{2} + a_1 \cos x + a_2 \cos 2x + \dots + b_1 \sin x + b_2 \sin 2x + \dots$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^0 f(x) dx + \frac{1}{\pi} \int_0^{\pi} f(x) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^0 x^2 dx + \frac{1}{\pi} \int_0^{\pi} 0 dx$$

861 page Exercise-11 $\rightarrow$ H.W

862 page Example-8 $\rightarrow$ H.W

Half-Range

$(0, \pi), (-\pi, \pi)$

Cosine
 $a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx ; b_n = 0$

Sine
 $b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx dx ; a_n = 0$

844 page Example-10

847 page Ex-12

871 Ex-17

872 page Ex-18

Fourier Analysis (14/01/24)
Parseval's Formula

$$\int_{-L}^L [f(x)]^2 dx = L \left\{ \frac{1}{2} a_0^2 + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \right\}$$

$$\boxed{\begin{array}{l} \text{Ex-20} \\ \text{p-875} \end{array}}$$

$$\boxed{\begin{array}{l} \text{H.W-20} \\ \text{p-876} \end{array}}$$

By using sine series for $f(x)=1, 0 < x < \pi$
show that $\frac{\pi}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots$

$$f(x) = \sum b_n \sin nx$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx dx = \begin{cases} \frac{2}{n\pi} & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$$

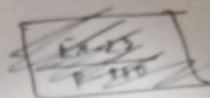
$$1 = \frac{4}{\pi} \sin x + \frac{4}{3\pi} \sin 3x + \dots$$

$$\int_0^{\pi} [f(x)]^2 dx = \frac{c}{2} [b_1^2 + b_2^2 + \dots]$$

$$\int_0^{\pi} dx = \frac{\pi}{2} \left[\left(\frac{4}{\pi} \right)^2 + \left(\frac{4}{3\pi} \right)^2 + \dots \right]$$

$$\frac{z}{s} = 1 + \frac{1}{s} + \frac{1}{s^2} + \dots$$

Fourier series in complex form



$$f(x) = \frac{a_0}{2} + a_1 \frac{\cos \pi x}{l} + a_2 \frac{\cos 2\pi x}{l} + \dots$$

$$b_1 \sin \frac{\pi x}{l} + b_2 \sin \frac{2\pi x}{l} + \dots$$

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}$$

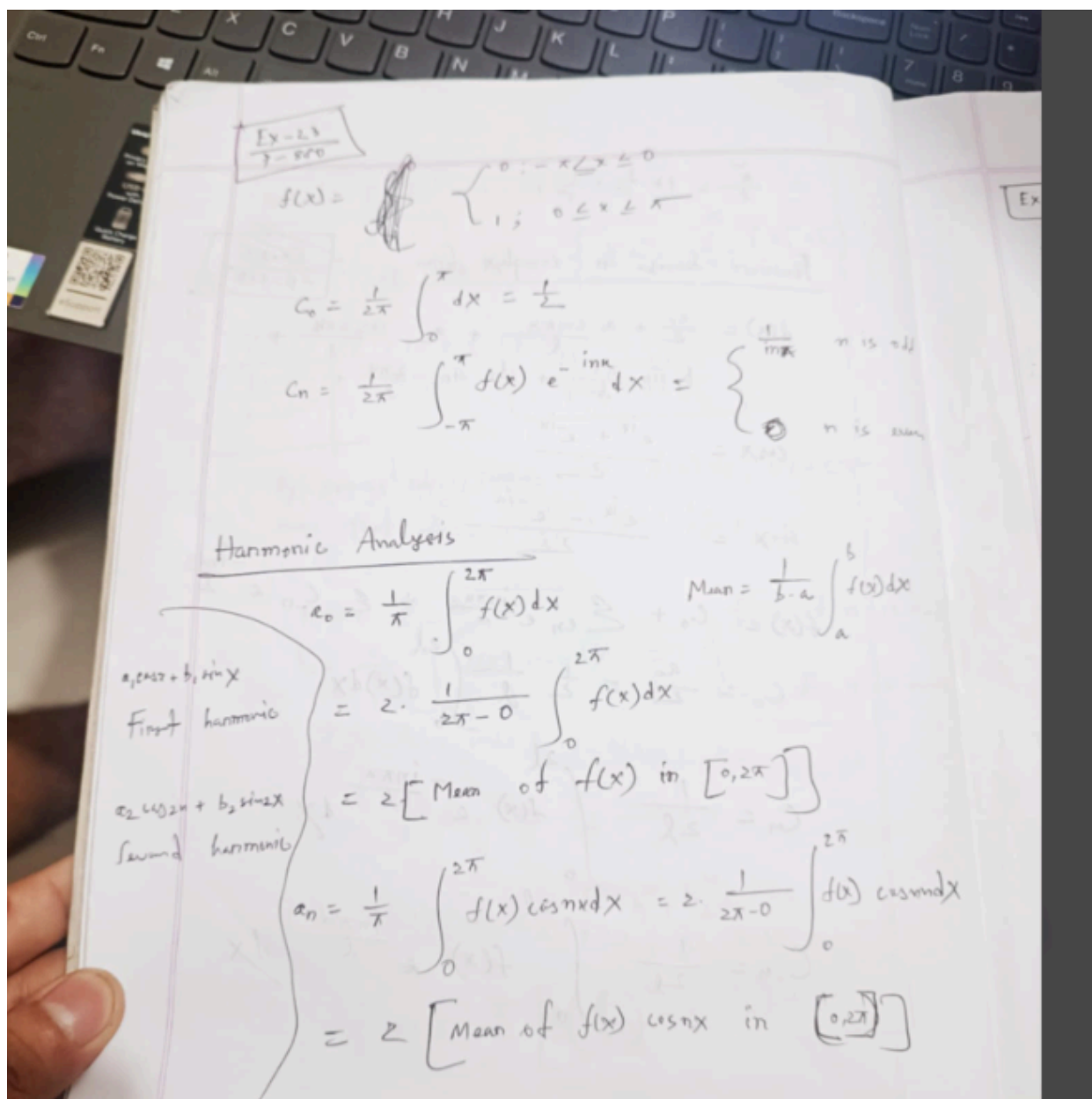
$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}$$

$$f(x) = C_0 + \sum C_n e^{\frac{i n \pi x}{l}} + \sum C_{-n} e^{-\frac{i n \pi x}{l}}$$

$$C_0 = \frac{a_0}{2} = \frac{1}{2l} \int_0^{2l} f(x) dx$$

$$C_n = \frac{1}{2l} \int_0^{2l} f(x) e^{-\frac{i n \pi x}{l}} dx$$

$$C_{-n} = \frac{1}{2l} \int_0^{2l} f(x) e^{\frac{i n \pi x}{l}} dx$$



Chapter 14 poracche antu ashe nai oidin.....

Note: samner assighment e fourirar e full clear hye jabe with questions

Sycology

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